Assignment 2

In [ ]:

1. What is a List?

ANS -

List is iterable object which store multiple heterogeneous data. it is mutable.
Ex - l = [1,2,3,"data"] , here l is a list

2. What is a Tuple?

Ans -

TUple is a iterable object which store multiple heterogeneous data. it is immutable.
Ex - t = (1,2,3,4,"python")

3. What is the difference between List and Tuple?

Ans -

1) list is mutable. tuple is immutable
2) list consume more memory. tuple consume less memory.
3) list take less time for computation but tuple consume more time

4. Python Program to find the largest element in the list.

In [4]:
```
l = list(input("Enter list"))
print(max(l))
```

6

5. Python program to interchange first and last elements in a list.

In [8]:
```
l = list(input("Enter list").split())
l[0],l[-1] = l[-1],l[0]
print(l)
```

```
['6', '2', '3', '4', '5', '1']
```

7. Python program to Reverse a List

In [9]:
```
l = list(input("Enter list").split())
l.reverse()
print(l)
```

```
['6', '5', '4', '3', '2', '1']
```

8. Python program to count occurrences of an element in a list

In [14]:
```
l = list(input("Enter list").split())
dic = {}
for i in l:
    dic[i] = dic.get(i,0) + 1
print(dic)
```

```
{'hello': 2, 'brother': 3}
```

9. Python program to find the sum of elements in a list

In [22]:
```
l= list(map(int , input("Enter list").split()))
print(sum(l))
```

```
21
```

10. Python program to Multiply all numbers in the list

In [23]:
```
l= list(map(int , input("Enter list").split()))
res = 1
for i in l:
    res *= i
print(res)
```

```
720
```

11. What are the ways to find the length of a list

Ans -

1) len function
2) loop and count variable - iterate list and increment count . count will be length of list

12. Python program to find the smallest and largest number in a list (Without min-max function)

In [1]:
```
import sys

l= list(map(int , input("Enter list").split()))
ma = - sys.maxsize -1
mi = sys.maxsize
for i in l:
    if ma < i:
        ma = i
    if mi > i:
        mi = i
print("smallest no -", mi)
print("largest no -", ma)
```

```
smallest no - 1
largest no - 6
```

13. Python Program to find the area of a circle

In [3]:
```
l = int(input("Enter radius"))
print(3.14 * l *l)
```

```
50.24
```

14. Take inputs from the user to make a list. Again take one input from the user and delete that element from a list.

In [4]:
```
l= list(input("Enter list").split())
v = input("Enter value need to be deleted")

l.remove(v)
print(l)
```

```
['1', '2', '3', '4', '5', '6']
```

15. You are given a list of integer elements. Make a new list that will store a square of elements of the previous list. (With and without list comprehension)

● Input_list = [2,5,6,12] ● Output_list = [4,25,36,144]

In [5]:
```
l= list(map(int , input("Enter list").split()))
res = []
for i in l:
    res.append(i*i)
print(res)
```

```
[1, 4, 9, 16, 25, 36]
```

16. WAP to create two lists, one containing all even numbers and the other containing all odd numbers between 0 to 151

In [8]:
```
even = []
odd = []
for i in range(0,152):
    if i % 2 == 0:
        even.append(i)
    else:
        odd.append(i)
print("even no list ",even)
print()
print("odd no list ",odd)
```

```
even no list  [0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 3
6, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 7
8, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 1
16, 118, 120, 122, 124, 126, 128, 130, 132, 134, 136, 138, 140, 142, 144, 146, 148,
150]

odd no list  [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37,
39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 73, 75, 77, 79,
81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117,
119, 121, 123, 125, 127, 129, 131, 133, 135, 137, 139, 141, 143, 145, 147, 149, 151]
```

17. Python program to Count Even and Odd numbers in a List

In [9]:
```
l= list(map(int , input("Enter list").split()))
even = 0
odd = 0
for i in l:
    if i % 2 == 0:
        even += 1
    else:
        odd += 1
print(f'even no - {even} odd no - {odd}')
```

```
even no - 5 odd no - 5
```

18. WAP to make new lists, containing only numbers which are divisible by 4, 6, 8, 10, 3, 5, 7, and 9 in separate lists for range(0,151)

In [18]:
```
dic = {}
for i in range(3,11):
    l = "list"+str(i)
    dic[l] = []
for i in range(0,152):
    for j in range(3,11):
        if i % j == 0:
            l = "list"+str(j)
            dic[l].append(i)

for i in dic:
    l = "list"+str(i)
    print(f"list of number divisible by {i} - {dic[i]}")
    print()
```

```
list of number divisible by list3 - [0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54, 57, 60, 63, 66, 69, 72, 75, 78, 81, 84, 87, 90, 93, 96, 99, 102, 105, 108, 111, 114, 117, 120, 123, 126, 129, 132, 135, 138, 141, 144, 147, 150]

list of number divisible by list4 - [0, 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72, 76, 80, 84, 88, 92, 96, 100, 104, 108, 112, 116, 120, 124, 128, 132, 136, 140, 144, 148]

list of number divisible by list5 - [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150]

list of number divisible by list6 - [0, 6, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, 72, 78, 84, 90, 96, 102, 108, 114, 120, 126, 132, 138, 144, 150]

list of number divisible by list7 - [0, 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84, 91, 98, 105, 112, 119, 126, 133, 140, 147]

list of number divisible by list8 - [0, 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104, 112, 120, 128, 136, 144]

list of number divisible by list9 - [0, 9, 18, 27, 36, 45, 54, 63, 72, 81, 90, 99, 108, 117, 126, 135, 144]

list of number divisible by list10 - [0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150]
```

19. Given a list containing elements of different types (integers, strings, and floats), create three separate lists to store integers, strings, and floats respectively.

In [6]:
```python
intl = []
floatsl = []
strl = []
l = [1 ,2 ,3 ,4, 5.5, 6.7, 9.1, "python", "java", 'ram']
for i in l:
    try:
        res = int(i)
        if res == i:
            intl.append(i)
        else:
            floatsl.append(i)
    except:
            strl.append(i)
print("int list ",intl)
print()
print("float list ",floatsl)
print()
print("str list ",strl)
print()
```

```
int list  [1, 2, 3, 4]

float list  [5.5, 6.7, 9.1]

str list  ['python', 'java', 'ram']
```

20. What's The Difference Between The Python append() and extend() Methods?

Ans-

append - it is used to insert parameter which is provided to it as element of list
extend - it is used to merge two list. as element of second list get inserted in first list as element

21. Write a Python program to append a list to the second list

In [7]:
```
l1 = [1,2,3,4,5,6]
l2 = [7,8,9,0]
l1.extend(l2)
print(l1)
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 0]
```

22. Write a Python program to find the third-largest number in a list

In [11]:
```
import sys
l= list(map(int , input("Enter list").split()))
first = - sys.maxsize -1
second = - sys.maxsize -1
third = - sys.maxsize -1
for i in l:
    if first < i:
        third = second
        second = first
        first = i
    elif first != i and second < i:
        third = second
        second = i
    elif first != i and second != i and third <i:
        third = i

print(third)
```

```
3
```

23. Write a Python program to check whether a list contains a sublist

In [28]:
```
l = list(input("enter list").split())
s = list(input("enter sublist").split())
index = -1
index2 = -1
n = 0
for i in s:
    if index == -1 and i in l:
        index = l.index(i)
        n = 1
    elif index != -1 and i in l:
        index2 = l.index(i)
        if index + 1 != index2:
            n = 0
            break
        else:
            index = index2
            n = 1
    else:
        n = 0
        break

if n == 0:
    print("not prsent")
else:
    print("present")
```

```
present
```

24. Write a Python program to generate all sublists of a list

In [36]:
```
l = list(input("enter list").split())
s = set()
s.add(tuple([]))
le = len(l)
for i in range(0,le):
    ls = []
    ls.append(i)
    s.add(tuple(ls))
    for j in range(i+1,le):
        ls.append(j)
        s.add(tuple(ls))
for i in s:
    print(list(i))
```

```
[0, 1]
[1, 2]
[2]
[3, 4]
[0, 1, 2]
[1, 2, 3]
[4]
[1, 2, 3, 4]
[1]
[2, 3]
[0, 1, 2, 3]
[0, 1, 2, 3, 4]
[2, 3, 4]
[0]
[]
[3]
```

25. Write a Python program to find common items from two lists

In [2]:
```
l = set(list(input("enter list 1 ").split()))
s = set(list(input("enter list 2 ").split()))
print(l.intersection(s))
```

```
{'4', '2'}
```

26. How to flatten a list in Python?

In [5]:
```
import numpy as np
l = [[1,2,3],[4,5,6]]
(np.array(l)).flatten()
```

Out[5]:
```
array([1, 2, 3, 4, 5, 6])
```

27. How to sort a list in ascending and descending order without using the sort function?

In [10]:
```
l = [1,2,4,3,4,5,6]

for i in range(1,len(l)):
    for j in range(i,0,-1):
        if l[j] < l[j-1]:
            l[j],l[j-1] = l[j-1],l[j]
print("ascending order - ",l)
l.reverse()
print("descending order - ",l)
```

```
ascending order -  [1, 2, 3, 4, 4, 5, 6]
descending order -  [6, 5, 4, 4, 3, 2, 1]
```

28. How to sort a tuple?

In [11]:
```
l = (1,2,4,3,4,5,6)
l = list(l)
for i in range(1,len(l)):
    for j in range(i,0,-1):
        if l[j] < l[j-1]:
            l[j],l[j-1] = l[j-1],l[j]
print("ascending order - ",tuple(l))
l.reverse()
print("descending order - ",tuple(l))
```

```
ascending order -  (1, 2, 3, 4, 4, 5, 6)
descending order -  (6, 5, 4, 4, 3, 2, 1)
```

29. Write a Python program to convert a list of multiple integers into a single integer

- [11, 33, 50] >>> 113350

In [12]:
```
l = list(input("Enter list").split())
s = "".join(l)
print(s)
```

```
12345
```

30. Difference between del and clear()

Both are used to delete elements, but they work differently.

del:

- Deletes a particular element using its index.
- Can delete multiple elements using slicing.
- Can delete the entire list variable.

clear():

- Removes all elements from the list.
- The list variable still exists as an empty list.

31. Difference between remove() and pop()

Both remove elements from a list.

remove():

- Removes an element based on its value.
- Removes the first occurrence of that value.
- Does not return the removed element.

pop():

- Removes an element based on its index.
- Returns the removed element.
- By default, removes the last element.

32. Difference between Indexing and Slicing

Indexing: Accesses a single element from a list.

Slicing: Accesses a range of elements from a list.

Both are used to arrange elements in ascending or descending order.

sort():

- Works only with lists.
- Modifies the original list.
- Returns None.

sorted():

- Works with any iterable (list, tuple, string, etc.).
- Returns a new sorted list.
- Does not modify the original iterable.

34. Difference between reverse() and reversed()

Both are used to reverse the order of elements.

reverse():

- List method.
- Modifies the original list.
- Returns None.

reversed():

- Built-in function.
- Does not modify the original list.
- Returns a reverse iterator.

35. Difference between Shallow Copy and Deep Copy

Both are used to create copies of objects.

Shallow Copy:

- Creates a new outer object.
- Nested objects are shared between the original and copied objects.
- Changes to nested mutable objects affect both lists.

Deep Copy:

- Creates a new outer object and recursively copies nested objects.
- Nested objects are independent.
- Changes to nested mutable objects do not affect the original.

36 How to check whether a list is empty or not?

1. Boolean evaluation - if not l:
2. Using length - if len(l) == 0:
3. Direct comparison -[if l == []:

37. How to concatenate two lists?

In [18]:
```
#Method 1: Using + operator
l1 = [10, 20, 30]
l2 = [40, 50, 60]
l3 = l1 + l2
print(l3)



#Method 2: Using extend()
l1 = [10, 20, 30]
l2 = [40, 50, 60]
l1.extend(l2)
print(l1)



#Method 3: Using unpacking
l1 = [10, 20, 30]
l2 = [40, 50, 60]
l3 = [*l1, *l2]
print(l3)
```

```
[10, 20, 30, 40, 50, 60]
[10, 20, 30, 40, 50, 60]
[10, 20, 30, 40, 50, 60]
```

38. Given a list of integers, create a new list where each element is replaced by the next largest number in the original list. If there is no larger number, replace the element with -1.

● input_list = [4, 2, 6, 8, 1, 2] ● output_list = [6, 6, 8, -1, 2, -1]

In [33]:
```
l = list(map(int, input("enter list").split()))
res = []

for i in range(0,len(l)):
    k = -1
    for j in range(i+1,len(l)):
        m = l[i:j+1]
        k = -1
        if m:
            if l[i] < max(m):
                k = max(m)
                break
    res.append(k)
print(res)
```

```
[6, 6, 8, -1, 2, -1]
```

39. Write a function to reverse the elements within each tuple in a list of tuples.

● tuples = [(1, 2), (3, 4), (5, 6)] ● Output: [(2, 1), (4, 3), (6, 5)]

In [35]:
```
l = [(1, 2), (3, 4), (5, 6)]
for i in range(0,len(l)):
    r = list(l[i])
    r.reverse()
    r = tuple(r)
    l.insert(i,r)
    l.remove(l[i+1])
print(l)
```

```
[(2, 1), (4, 3), (6, 5)]
```

40. Write a function to add corresponding elements of tuples in two lists of tuples

● list1 = [(1, 2), (3, 4)] ● list2 = [(5, 6), (7, 8)] ● Output: [(6, 8), (10, 12)]

In [36]:
```
list1 = [(1, 2), (3, 4)]
list2 = [(5, 6), (7, 8)]
res = []
for i in range(0,len(list1)):
    l1 = list1[i]
    l2 = list2[i]
    add = []
    for j in range(0,len(l1)):
        add.append(l1[j] + l2[j])
    add = tuple(add)
    res.append(add)
print(res)
```

```
[(6, 8), (10, 12)]
```

In [ ]: